

AUTOSAR High-Level Design: Electronic Braking System

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1. Architecture Overview

The Electronic Braking System contains three core software components:

- SWC_BrakeControl: Executes driver deceleration requests and ABS regulation.
- SWC_SensorFusion: Consolidates raw wheel speeds and yaw rate sensor inputs.
- SWC_ActuatorInterface: Interfaces with electromechanical hydraulic valves.

2. Ports and Interface Mapping

SWC_BrakeControl specifies the following communication endpoints:

- Required Port: Rp_WheelPulse with Interface IVehicleSpeed (from SWC_SensorFusion).
- Provided Port: Pp_DecelerationCmd with Interface IBrakeActuation (to SWC_ActuatorInterface).
- Client-Server Port: Cp_Diagnostics with Interface IDiagnosticControl.

Architectural Rule: If IVehicleSpeed signals fail, SWC_BrakeControl shifts to fail-safe within 15 ms.